

Data Link, physical

Network Access. layer

- 4NF it must satisfy two conditions:
- 1. it should be in BCNF.
 - 2. it should not have Multi-valued Dependency.
- what is Multi-valued Dependency?

A dependency:

$A \twoheadrightarrow B$ is Multi-val-D. if for single value of A, more than one value of B exist in a table.

and A table should have at least 3 columns to have Multi-valued Dependency in a table.

But why?

Because if we have only two columns in table.

A	B
A ₁	B ₁
	B ₂

we can simply distribute the Multivalue data in multiple rows and there is no need for decomposition the table.

A	B
A ₁	B ₁
A ₁	B ₂

For a table with A, B, C columns.

$A \rightarrow B \rightarrow$ is Multi-values Dep

Then B and C should be independent of each other

So there are three condi. for Mult-VDep:

- $A \rightarrow B$, for a single value of A more than one value of B exist.

Table should have at least 3 columns

- For this table with A, B, C columns, B and C should be independent.

Note:

Multi-value Dep occurs due to bad data base design.

ex:

The problem (violating 4NF):-

Imagine tracking a developer's skills and languages spoken. Both depend on the developer.

But a person's coding skills and spoken languages have zero relationship with each other.

if you put them into a single table, you are going to store every combination

Developer	skill	language spoken
Sarah	python	English
Sarah	python	spanish
Sarah	SQL	English
Sarah	SQL	spanish

- why this is broken: if sarah learns a 3rd skill (eg. Docker), you must insert 2 brand new rows just to pair Docker with both English and spanish. if she has 10 skills and 5 lang. you would need $10 \times 5 = 50$ rows

The solution (Achieving 4NF).

To Achieve 4NF, split the independent multi-val facts into two separate tables

Developer-skills

Developer	Skills
Sarah	python
Sarah	SQL

Developer-Languages

Developer
Sarah
sarah

langSpoken.
English
Spanish.

- why this works Now, adding a new skill for Sarah requires adding just 1 row in Developer-Skills without touching Developer-Languages

Quick rule of thumb

- 1NF to BCNF prevents single columns from holding inaccurate or dependent data.

• 4NF stops two independent multi-valued lists from being crammed into the same data